

- Description

This module manages a battery charging system, ensuring that:

1. **The battery charge level is continuously monitored and maintained within a safe range.**
2. **If the charge level exceeds the maximum threshold, an overcharge warning is triggered.**
3. **If the charge level drops below the minimum threshold, a low-battery alert is issued.**
4. **A log of all charge level readings, alerts, and charging actions is maintained for battery lifespan optimization.**

This system helps prevent overcharging and deep discharge, thereby extending battery life and ensuring safe operation.

- Specification

module BatteryChargingManagement

var currentCharge, maxCharge, minCharge: real

var chargeLog: sequence of string

process updateCharge(changeAmount: real)

functional scenario form:

currentCharge $\geq$ minCharge && chargeLog = chargeLog $\oplus$ ("Charge Level Updated: " + currentCharge + "%"), Charge level updated successfully.

||

currentCharge > maxCharge && chargeLog = chargeLog $\oplus$ ("Overcharge Warning: " + currentCharge + "%, Stop Charging Immediately."), Overcharge warning triggered, stop charging.

||

currentCharge < minCharge && chargeLog = chargeLog $\oplus$ ("Low Battery Alert: " + currentCharge + "%, Immediate Charging Recommended."), Low battery alert triggered, immediate charging recommended.

end process

end module

- Java Code

```
package code;
import java.util.*;
class Battery {
    private double maxCharge;
    private double minCharge;
    private double currentCharge;
    private List<String> chargeLog;
    public Battery(double maxCharge, double minCharge, double initialCharge) {
        this.maxCharge = maxCharge;
```

```

        this.minCharge = minCharge;
        this.currentCharge = initialCharge;
        this.chargeLog = new ArrayList<>();
    }
    public void updateCharge(double changeAmount) {
        currentCharge += changeAmount;
        String message = String.format("Current Charge Level: %.1f%%",
currentCharge);
        if (currentCharge > maxCharge) {
            message += " | WARNING: Overcharge Detected! Stop Charging
Immediately.";
        } else if (currentCharge < minCharge) {
            message += " | ALERT: Low Battery! Immediate Charging Recom
mended.";
        }
        System.out.println(message);
        chargeLog.add(message);
    }
    public void printChargeLog() {
        System.out.println("\nBattery Charge Log:");
        for (String log : chargeLog) {
            System.out.println(log);
        }
    }
}

public class c34_BatteryManagementSystem {
    public static void main(String[] args) {
        Battery battery = new Battery(100.0, 20.0, 50.0);
        // testcase-VT:
        //     battery.updateCharge(30.0);
        //     battery.updateCharge(-40.0);
        //     battery.updateCharge(-25.0);
        //     battery.updateCharge(60.0);
        //     battery.updateCharge(35.0);

        // testcase-FT:
        //     battery.updateCharge(-1.7976931348623157E308);
        //     battery.updateCharge(1.7976931348623157E308);
        //     battery.updateCharge(2.0444353298434178E307);
        //     battery.updateCharge(0);
        //     battery.updateCharge(-4.9E-324);

        // testcase-VT:
        //     battery.updateCharge(3.0);
    }
}

```

```
//      battery.updateCharge(4.0);
//      battery.updateCharge(2.0);
//      battery.updateCharge(6.0);
//      battery.updateCharge(3.0);
//
//      // testcase-VT:
//      battery.updateCharge(3);
//      battery.updateCharge(-4);
//      battery.updateCharge(-2);
//      battery.updateCharge(600);
//      battery.updateCharge(3);
//
//      battery.printChargeLog();
//  }
}
```

- Coverage report

1. Test case number: 5
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.7 %	6	836	842
▼ src	0.7 %	6	836	842
▼ code	0.7 %	6	836	842
▼ c34_BatteryManagementSystem.java	100.0 %	6	0	6
> Battery	100.0 %	6	0	6
> c34_BatteryManagementSystem		0	0	0

3. Vibration testing instruction coverage: 97.2%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.6 %	106	19,099	19,205
▼ src	0.6 %	106	19,099	19,205
▼ code	0.6 %	106	19,099	19,205
▼ c34_BatteryManagementSystem.java	97.2 %	106	3	109
> c34_BatteryManagementSystem	89.3 %	25	3	28
> Battery	100.0 %	81	0	81

4. Fuzz testing branch coverage: 83.3%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.6 %	5	837	842
▼ src	0.6 %	5	837	842
▼ code	0.6 %	5	837	842
▼ c34_BatteryManagementSystem.java	83.3 %	5	1	6
> Battery	83.3 %	5	1	6
> c34_BatteryManagementSystem		0	0	0

5. Fuzz testing instruction coverage: 97.2%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.6 %	106	19,099	19,205
▼ src	0.6 %	106	19,099	19,205
▼ code	0.6 %	106	19,099	19,205
▼ c34_BatteryManagementSystem.java	97.2 %	106	3	109
> c34_BatteryManagementSystem	89.3 %	25	3	28
> Battery	100.0 %	81	0	81

6. Z3 branch coverage: 33.3%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.2 %	2	838	840
▼ src	0.2 %	2	838	840
▼ code	0.2 %	2	838	840
▼ c34_BatteryManagementSystem.java	33.3 %	2	4	6
> Battery	33.3 %	2	4	6
> c34_BatteryManagementSystem		0	0	0

7. Z3 instruction coverage: 72.9%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.3 %	78	22,312	22,390
▼ src	0.3 %	78	22,312	22,390
▼ code	0.3 %	78	22,312	22,390
▼ c34_BatteryManagementSystem.java	72.9 %	78	29	107
> Battery	67.9 %	55	26	81
> c34_BatteryManagementSystem	88.5 %	23	3	26

8. Unimproved V-method branch coverage: 50%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.4 %	3	837	840
▼ src	0.4 %	3	837	840
▼ code	0.4 %	3	837	840
▼ c34_BatteryManagementSystem.java	50.0 %	3	3	6
> Battery	50.0 %	3	3	6
> c34_BatteryManagementSystem		0	0	0

9. Unimproved V-method instruction coverage: 76.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	82	22,308	22,390
▼ src	0.4 %	82	22,308	22,390
▼ code	0.4 %	82	22,308	22,390
▼ c34_BatteryManagementSystem.java	76.6 %	82	25	107
> Battery	72.8 %	59	22	81
> c34_BatteryManagementSystem	88.5 %	23	3	26